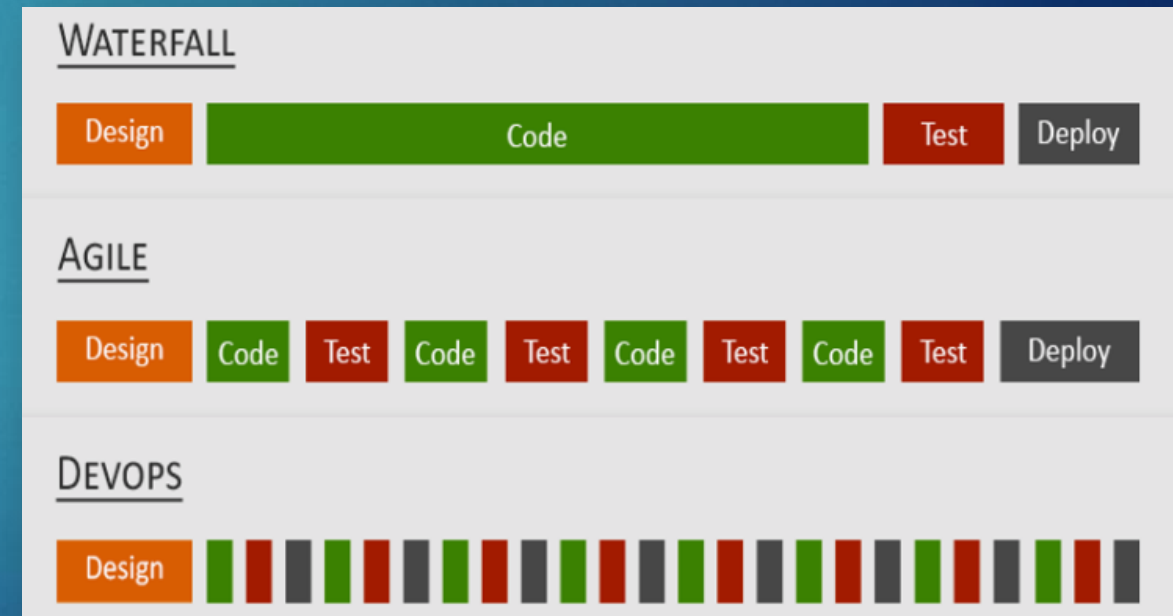
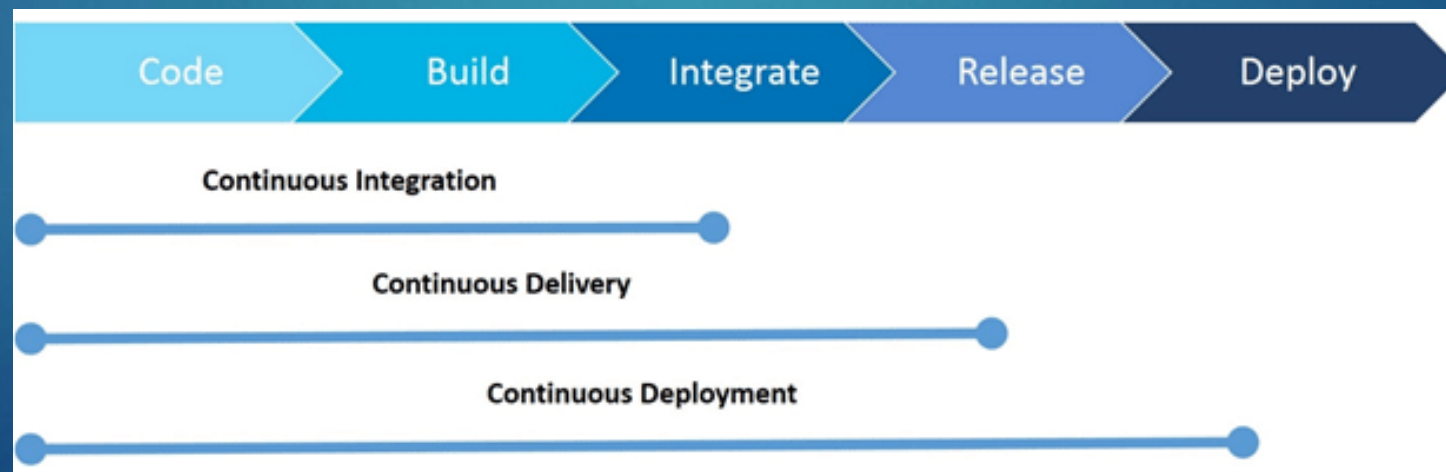


- ▶ DevOps is a set of practices that combines software development (Dev) and information-technology operations (Ops) which aims to shorten the systems development life cycle and provide continuous delivery with high software quality
- ▶ It can be thought of as a branch of Agile
- ▶ It ensures that there is an extra focus on the development and operations teams collaborating closely, especially during the release, deployment and production support phases of the SDLC
- ▶ DevOps is a cultural shift



- ▶ A CI/CD Pipeline is the backbone of the modern DevOps environment. It bridges the gap between development and operations teams by automating the building, testing, and deployment of applications:
 - ▶ **Continuous Integration (CI)** = short-lived feature branches, team is merging to master branch multiple times per day, fully automated build and test process; deployment is manual
 - ▶ **Continuous Delivery (CD)** = CI + the entire software release process is automated, it may be composed of multiple stages, and deployment to production is manual
 - ▶ **Continuous Deployment** = CI + CD + fully automated deployment to production



- ▶ GitHub Actions enables the creation of automated workflows directly within your GitHub repository, supporting CI/CD processes
- ▶ Triggering Workflows
 - ▶ Actions are initiated by GitHub events such as code pushes, pull requests, or scheduled events, allowing seamless automation in response to changes
- ▶ Deployment Environments
 - ▶ Workflows can run on GitHub-hosted virtual machines or custom self-hosted runners, providing flexibility to match your deployment needs
- ▶ Integration with Terraform
 - ▶ Combine GitHub Actions with Terraform to automate infrastructure provisioning alongside your code deployment, streamlining your DevOps pipeline



Understanding Cloud Costs

Engineering Cloud Software Systems

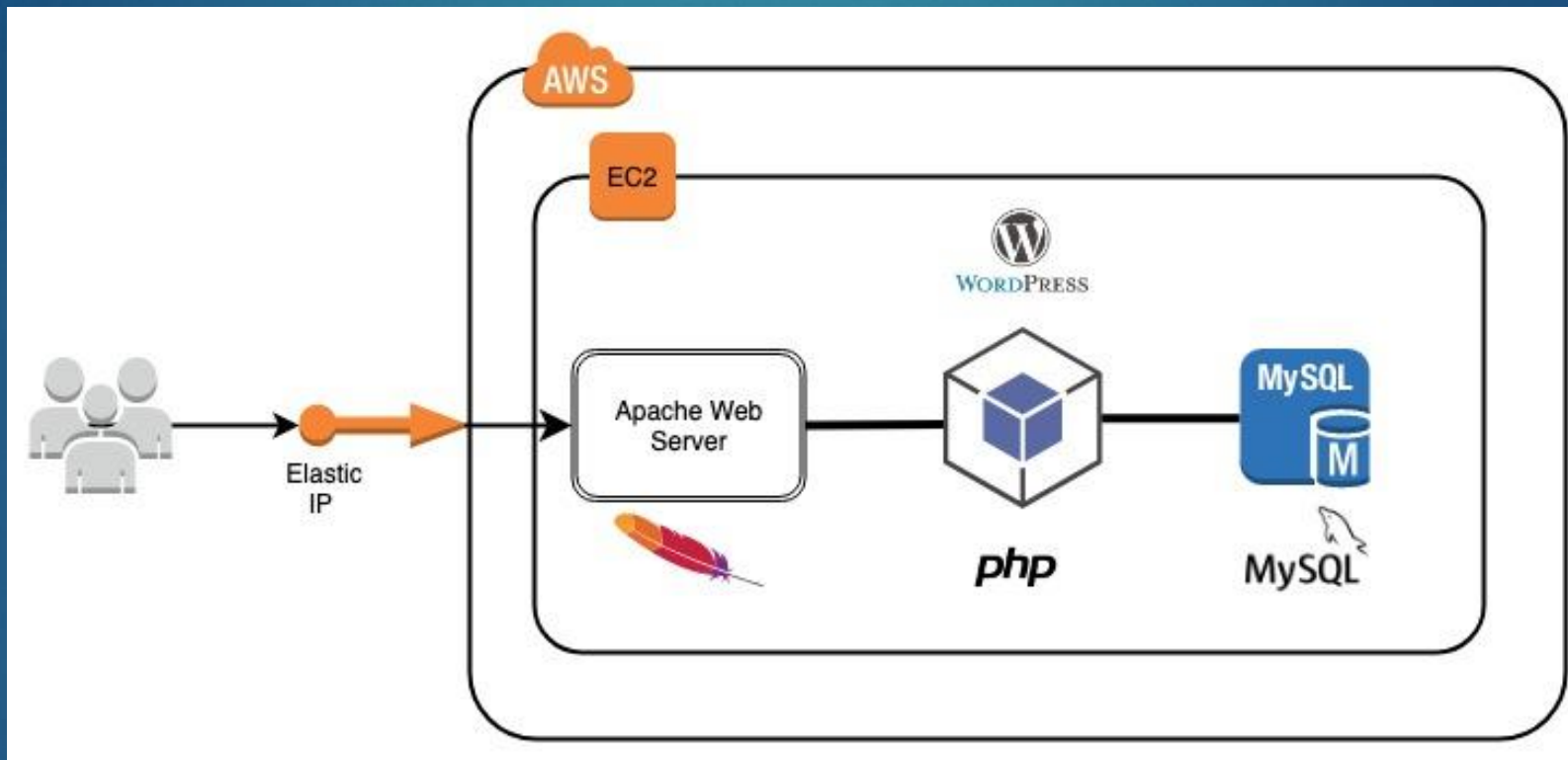
Department of Software Engineering
Rochester Institute of Technology



WordPress and EC2

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- ▶ You all have created a simple EC2 instance running a WordPress application



- ▶ When it comes to performance, scalability and reliability, what are some of the limitations you see with this application?

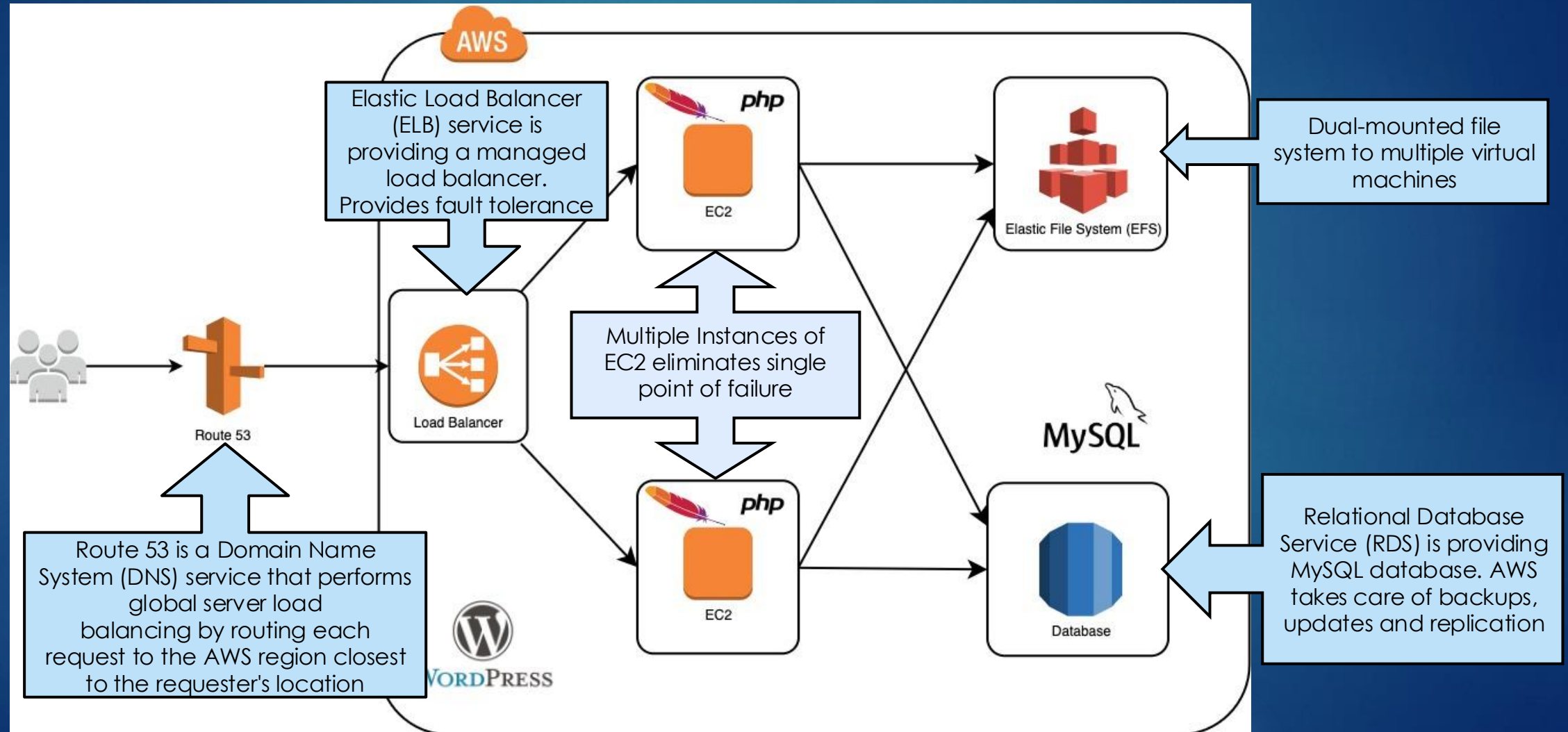
Some Noted Limitations

- ▶ Everything resides on a single server
 - ▶ All running processes share the same resources
- ▶ If anything crashes, app is unavailable (single point of failure)
- ▶ If anything needs to be updated (e.g. PHP, OS), the app will need to be taken offline
- ▶ No load balancing if there is problem accessing a region or single server that might be unavailable
- ▶ No external storage to expand over time
- ▶ No ability to expand as user demand increase
- ▶ How much do you think this application cost to run per month?
 - ▶ \$8.50/month



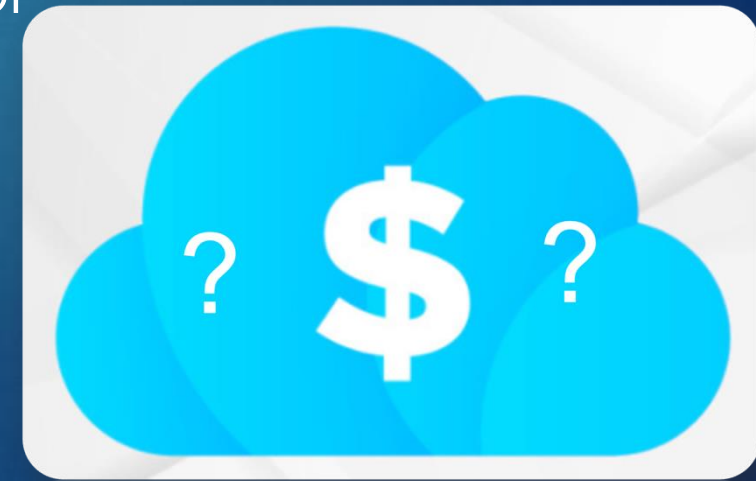
A Better Solution...

7



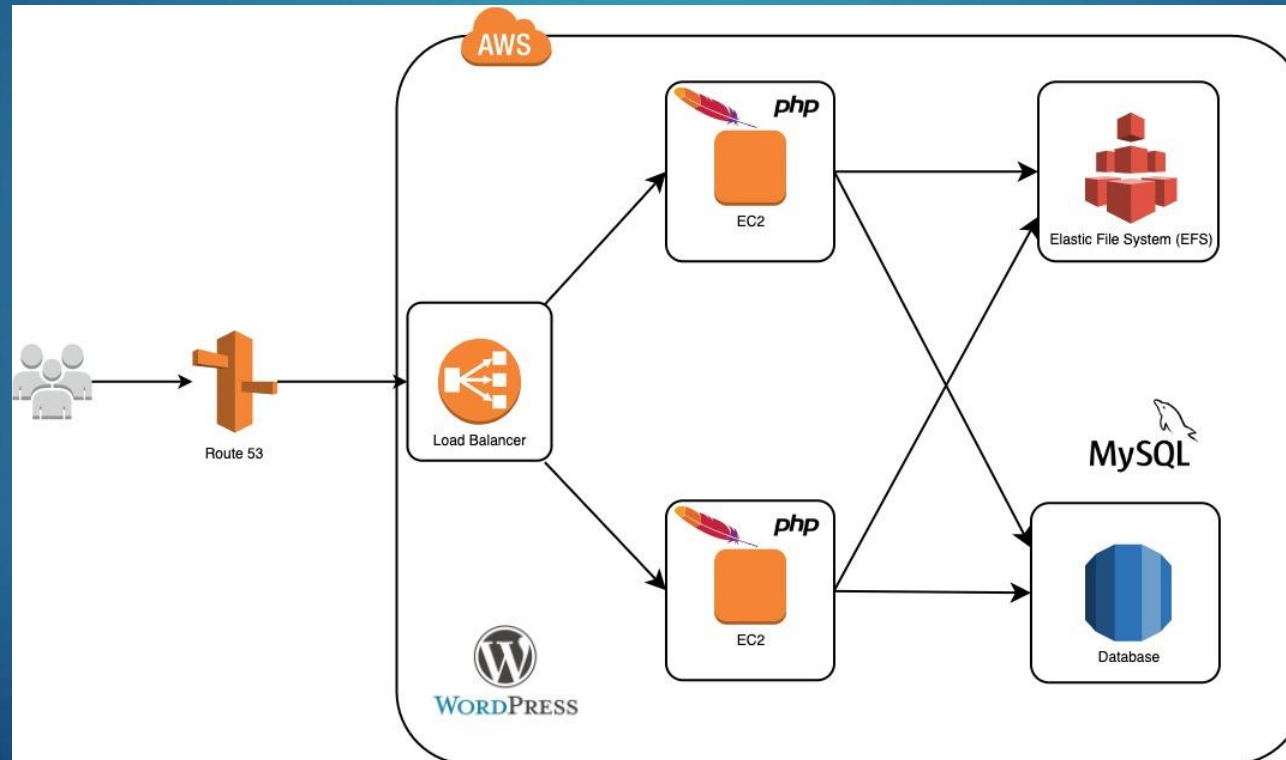
How to Estimate the Cost?

- ▶ There are many factors to consider when determining an estimate
 - ▶ How many requests are coming in per month?
 - ▶ What is the estimated amount of data to transfer in/out per month?
 - ▶ How much storage is needed?
 - ▶ What region is this running in?
 - ▶ What type of Support Model is needed?
- ▶ AWS has a solution called AWS Pricing Calculator
- ▶ Quick Tutorial: [Video](#)



How to Determine the Cost?

- ▶ Let's try an example by estimating the new WordPress solution below
 - ▶ Do the activity in [Assignments > Activity - AWS Calculator Exercise](#)
 - ▶ Make sure you take a screenshot when you are done



Understanding Cloud Costs

- ▶ More than 90% of organizations will use public cloud services this year, fueled by record cloud computing growth
- ▶ Public cloud customers will spend more than \$65 billion on Infrastructure as a Service (IaaS)
- ▶ Not understanding costs and how to control them can easily lead to “cloud waste”

TOP CHALLENGES FOR MANAGING CLOUD BUDGETS

44%

Difficulty determining best-fit workloads for public, private, hybrid cloud

39%

Difficulty planning and allocating budget for cloud consumption

37%

Lack of visibility of used services at the cost centre, workload and application level

Cloud Waste

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- ▶ These estimates only reflect costs associated with IaaS
- ▶ It does not include wasted spend on Platform as a Service (PaaS), which makes up \$55 billion in cloud spend according to Gartner's estimates

Cloud Waste - Examples

► Idle Resources

- Development servers are left running overnight and on weekends when they're not needed
- 44% of compute spend is tied to non-production environments. If these are only needed during a 40-hour work week, they sit idle 76% of the time representing a potential ~34% total compute cost reduction if powered down outside business hours.

► Overprovisioned Resources

- About 40% of instances are sized at least one size larger than needed for their workloads
- The cost can be cut in half by reducing an instance by one size, while downsizing by two sizes saves 75%

► Orphaned Volumes and Snapshots

- These are resources that have been detached from the infrastructure (e.g. EC2) they were created to support, such as a volume detached from an instance or a snapshot with no volume attachment



Another Cost Factor - Support

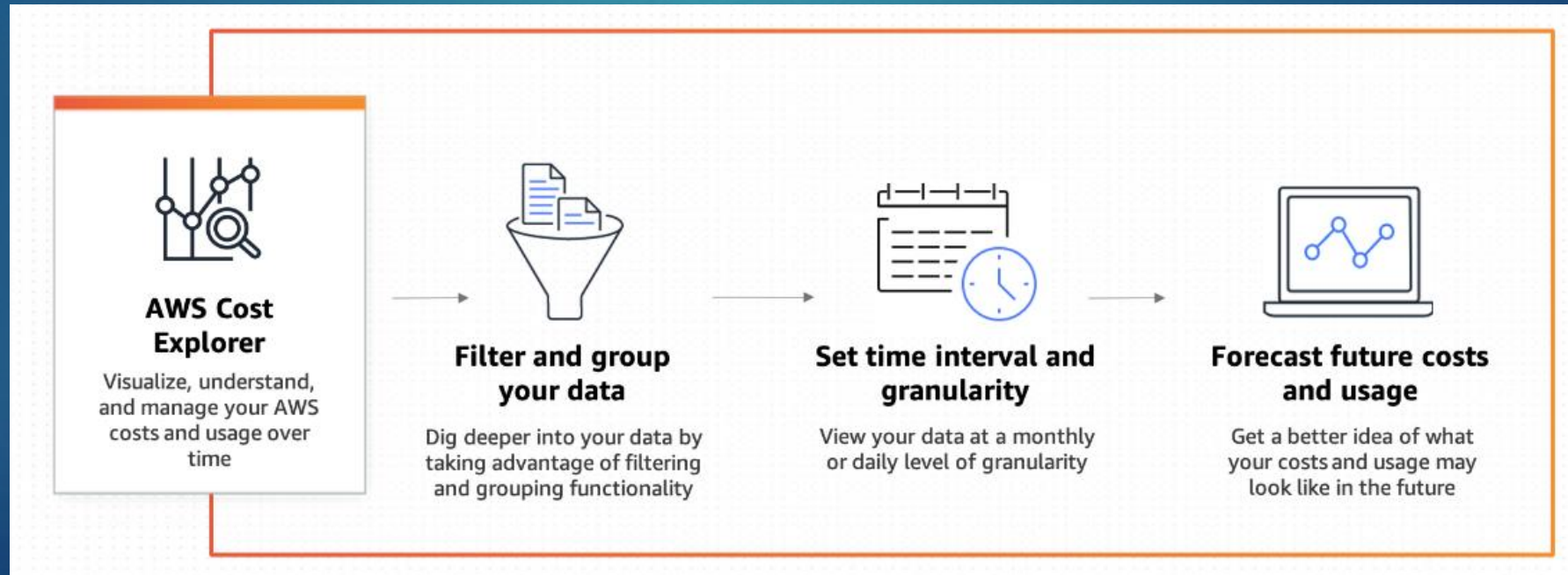
► AWS Support Plans

	BASIC	DEVELOPER	BUSINESS	ENTERPRISE
Cost	Free	\$29/mo	\$100/mo	\$15,000/mo
Use Case		Experimenting	Production use	Mission-critical use
Tech Support	NO	Business hour via e-mail	24x7 via email, chat & phone	24x7 via email, chat & phone
SLA		12-24 hrs at local business hours	1 hr response to urgent support cases	15 min to critical support cases w/ priority
TAM & Support Concierge	NO	NO	NO	YES
Support Cases	None	1 Person, Unlimited Cases	Unlimited contacts/cases	Unlimited contacts/cases

AWS Cost Explorer - Understanding your costs

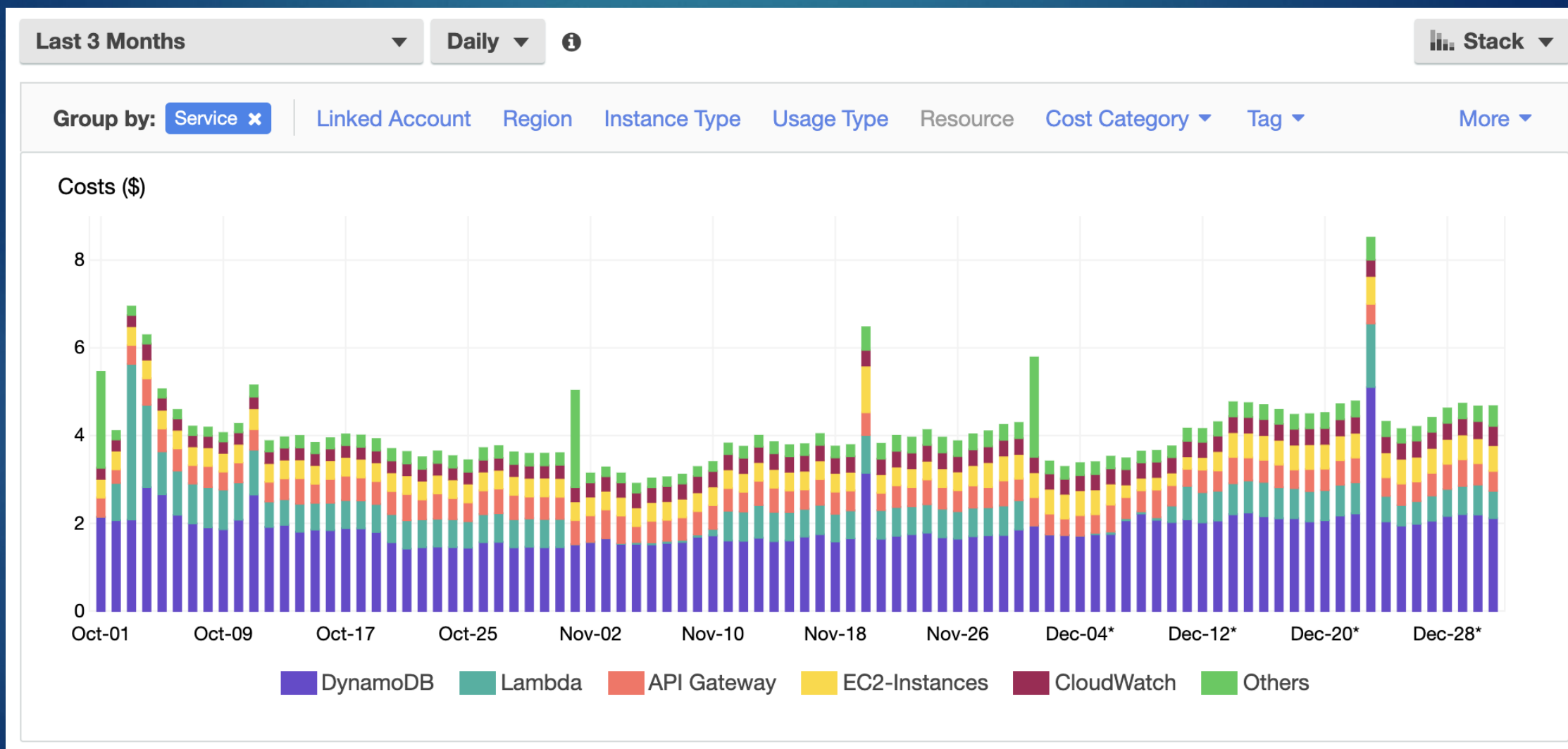
14

- ▶ Once you are running on the cloud, AWS Cost Explorer lets you visualize, understand, and manage your AWS costs and usage over time
 - ▶ Analyze your data at a high level (for example, total costs and usage across all accounts) or dive deeper into your cost and usage data to identify trends, pinpoint cost drivers, and detect anomalies



AWS Cost Explorer - Example

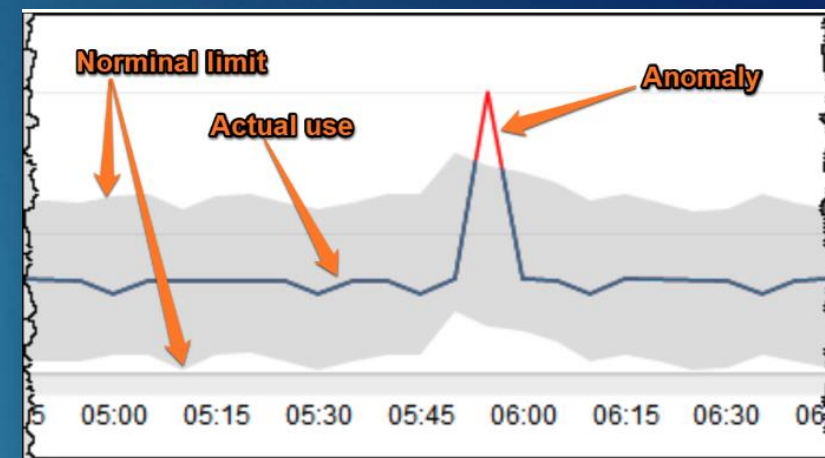
15



Cost Anomaly Detection

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- ▶ With Cost Anomaly Detection, AWS helps its users reduce costly surprises by identifying unexpected costs and their root causes in advance, so they can take action to avoid exceeding their budgets
- ▶ You'll receive alerts to any sign of an anomaly as soon as it appears



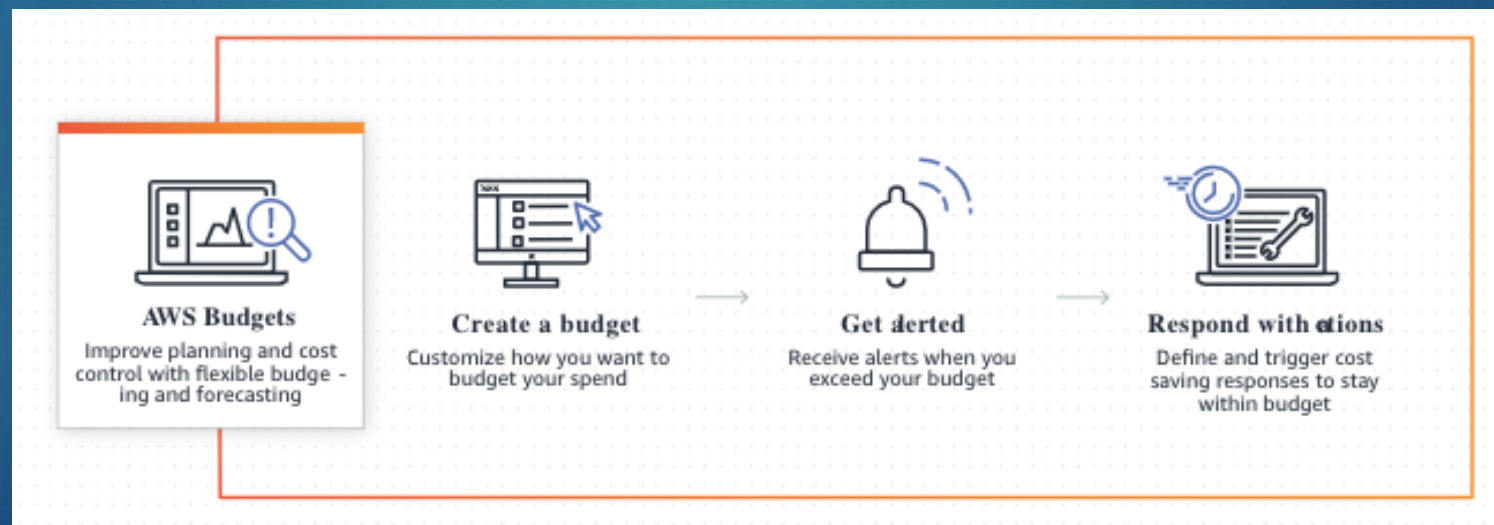
Monitor types

How to monitor your spend

- ☒ **AWS Services - Recommended**
This monitor evaluates each of the services you use individually, allowing smaller anomalies to be detected. Anomaly thresholds are automatically adjusted based on your historical service spend patterns.
- ☐ **Linked Account**
This monitor evaluates total spend for an individual Linked Account. This monitor can be helpful if your organization defines teams (or products, services, environments) by linked account.
- ☐ **Cost Category**
This monitor evaluates total spend for an individual cost category value. This monitor can be helpful if your organization defines teams (or products, services, environments) by using cost categories.
- ☐ **Cost Allocation Tag**
This monitor evaluates total spend for an individual tag key-value pair. This monitor can be helpful if your organization defines teams (or products, services, environments) by using cost allocation tags.

AWS Budgets – Set a spending plan

- ▶ With AWS Budgets, you can set custom spending plans on AWS for a specific period
 - ▶ You will also receive an alert if your AWS costs or usage exceeds or is projected to exceed your set spending limit
 - ▶ Example: Send me an email alert when I am within 80% of my weekly budget of \$20.00



AWS Budgets – Set a spending plan

- ▶ Let's set up a plan for our Free Tier accounts
- ▶ Go to AWS Budgets and click “Create a budget”

AWS Billing

AWS Budgets

Set custom budgets that alert you when you exceed your budgeted thresholds

AWS Budgets is your hub for creating, tracking, and inspecting your budgets.

Start tracking your AWS costs and usage

Once you have a budget created, AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.

Create a budget

AWS Budgets – Set a spending plan

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Budget setup

☒ **Use a template (simplified)**
Use the recommended configurations. You can change some configuration options after the budget is created.

☐ **Customize (advanced)**
Customize a budget to set parameters specific to your use case. You can customize the time period, the start month, and specific accounts.

Templates - new
Choose a template that best matches your use case.

☒ **Zero spend budget**
Create a budget that notifies you once your spending exceeds \$0.01 which is above the AWS Free Tier limits.

☐ **Monthly cost budget**
Create a monthly budget that notifies you if you exceed, or are forecasted to exceed, the budget amount.

☐ **Daily Savings Plans coverage budget**
Create a coverage budget for your Savings Plans that notifies you when you fall below the defined target.

☐ **Daily reservation utilization budget**
Create a utilization budget for your reservations that notifies you when you fall below the defined target.

Keep the default settings if you want to be alerted to any costs

Budget name
Provide a descriptive name for this budget.

Names must be between 1-100 characters.

Email recipients
Specify the email recipients you want to notify when the threshold has exceeded.

Maximum number of email recipients is 10.

Scope
All AWS services are in scope in this budget.

Notification
You will be notified via email when any spend above \$0.01 is incurred.

Template settings
This template has default configurations that can be changed later. To change any of these settings, see [Custom](#). You can also download this template in [JSON](#).

Enter email(s) you want to be alerted to and click “Create budget”. That’s it.



- ▶ As a consulting team, you need to provide a very high-level estimate to an organization for a particular scenario they are looking to solve
- ▶ You will use the AWS Pricing Calculator to put together your estimate
- ▶ Each team will be assigned a scenario, which describes the task, but details might be missing
- ▶ Your job:
 - ▶ Define all the hardware you think is needed to architect the solution
 - ▶ Provide an estimate of the costs
 - ▶ Each team will share their estimates and assumptions made



- ▶ What will you need to consider for this exercise?
 - ▶ Computing power (e.g. servers)
 - ▶ Infrastructure Location (specific regions)
 - ▶ Storage (S3, EBS, EFS)
 - ▶ Bandwidth Allocation (how fast performance)
 - ▶ Service Level Agreements (SLA)/uptime requirements?
 - ▶ Support plan?



- ▶ Open [Assignments > Activity - Cost Estimating Scenarios](#) to retrieve details of your team scenario
- ▶ Download the template “Cost Estimating Scenario Template” from the same location
- ▶ This will not be perfect so when in doubt, make your best estimate
- ▶ Submit completed template and the estimate worksheet (CSV) to [Assignments > Activity - Cost Estimating Scenarios](#)
 - ▶ Use the AWS Pricing Calculator and use the “Export” to export your cost estimates
 - ▶ Document with all assumptions made
 - ▶ Only 1 submission per team
- ▶ We will share these with the class
 - ▶ Pick 1 person to be your spokesperson